CLOSE TRIPLE APPROACHES AND ESCAPE IN THE THREE-BODY PROBLEM

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Abstract. We have studied a total of 5000 close triple approaches resulting in escape, for equal-mass systems with zero initial velocities. Escape is shown to take place in the majority of the cases after a 'fly-by' close triple approach when the escaper passes near the centre of mass along an almost straight-line orbit. A number of configurational and kinematical parameters are introduced in order to characterize the triple approach. The distributions of these parameters are investigated. A comparison with 831 examples in the vicinity of the so-called Pythagorean problem is carried out. We find that the general features of close triple approaches which result in escape are the same for both types of systems.

Key words: Three-body problem, computer simulations, close triple approach, escape

1. Introduction

Computer simulations of the general three-body problem have resulted in a classification of the motions (Agekian and Anosova, 1967; Szebehely, 1971; Anosova, 1988). This classification was modified by Agekian and Martynova (1973) who added an important feature. The types of motion are as follows:

0. Close triple approach
1. Simple interplay
2. Ejection with return
3. Escape
4. Stable revolution
5. Lagrangian equilibrium configuration
6. Collisions and periodic orbits

Following the historical tradition, the state of close triple approach precedes the original classification. During the dynamical evolution of non-hierarchical triple systems, the states 0 – 2 can be interchanged, while the state 3 is final. Some criteria to distinguish the states 0 – 3 for planar non-rotating triple systems were suggested by Agekian and Martynova (1973). These criteria provide a possibility to separate

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the states 0, 1 and 2 combined with 3. In order to separate the latter, one may use various escape criteria which have been reviewed and analysed by Orlov (1986). The states 4 – 6 may be investigated both analytically and numerically (see e.g. the books of Golubev and Grebenikov, 1985, and Marchal, 1990).

A study of close triple approaches (Anosova and Zavalov, 1981, 1989; Szebehely, 1979 and references therein) showed that they are necessary but not sufficient conditions for escape. Statistically, the closer the triple approach, the more probable that an escape will occur. However, the closeness is not a unique criterion for escape.

Anosova and Zavalov (1981) separated the triple approaches into two basic categories: (I) 'Fly-by', when one of the components approaches the centre of mass, it increases the velocity and may escape; and (II) 'exchange', consisting of successive two-body approaches. In case (I), the trajectory of the escaper is usually a straight line, whereas in case (II) the trajectories of all bodies are strongly curved. Triple approaches of type (I) result in escape more often than those of type (II). Typical examples of approaches of both types are shown in Figure 4 of Anosova (1991), as well as in Anosova and Orlov (1992).

Some basic properties of close triple approaches of type (I) which result in escape were formulated by Agekian and Anosova (1991): (i) the triple approach must have a small perimeter; (ii) at the time t_0 when the escaper approaches the centre of mass, the two other bodies must recede; and (iii) the direction of motion of the escaper at time t_0 must be nearly orthogonal to the line connecting the two other bodies. The kinetic energy of the escaper is larger when the triple approach is closer, if the above orthogonality condition is satisfied and the escaper passes near the centre of mass. Initial conditions at the time of the closest triple approach were specified for the case of all three components on the same straight line.

In the present work, we do not introduce any limitation on the type of motion at the triple approach. We consider a sample of 5000 initial conditions for equal-mass components and zero velocities. We have also studied 831 examples of mass ratios 3:4:5 for configurations in the vicinity of the Pythagorean problem. Some general results of the former and latter cases are presented elsewhere (Anosova, Orlov and Aarseth, 1993; Aarseth *et al.*, 1994). In Section 2, we describe briefly the method of selecting initial conditions and the parameters characterizing triple approaches. The results of the statistical study are discussed in Section 3.

2. Initial Conditions and Triple Approach Parameters

We consider the dynamical evolution of triple systems with zero initial velocities. Three methods for generating the masses and initial coordinates have been used: (i) all masses are equal and the position of the third body is chosen at random within a segmented region D which consists of the positive quadrant in the $x - y$ plane within a circle centred on the first body fixed at $x = -0.5, y = 0$, whereas the second body is placed at $x = 0.5, y = 0$ (Figure 1, Anosova, 1986); (ii) the masses

are equal and all the positions are distributed uniformly within a circle R of unit radius; and (iii) the masses have ratios 3:4:5 and the initial positions are generated in the vicinity of the Pythagorean triangle (Aarseth *et al.*, 1994).

The dynamical evolution of every triple system was studied until the time when the Standish (1971) escape criterion was satisfied, or conditional escape (Anosova, 1986) occurred. The latter is defined when the distance of the third body exceeds $10\,d$ with respect to the remaining binary, where d is the virial radius; $d = -G(M_1M_2 + M_1M_3 + M_2M_3)/(2E)$, where G is the gravitational constant, M_k ($k = 1,2,3$) are the masses and E is the total energy. Here we use the same numerical algorithm as before.

In the present work, we concentrate on the triple approaches resulting in escape or conditional escape. We define a triple approach at the time of minimum moment of inertia, provided the corresponding perimeter (sum of the three separations) is smaller than 2 d. Hence a small proportion of the examples may not be included in the analysis. The following parameters are used to describe the triple approach:

- The time T of the last triple approach
- The minimum perimeter P achieved during the approach
- The minimum moment of inertia, I
- The impact parameter Y with respect to the centre of mass
- The deviation from the isosceles triangle: $Z = |r_1 - r_2|/(r_1 + r_2)$
- The angle Ψ between the velocity vector of the escaper and the straight line connecting the two other components
- The relative velocity V_b of the radial motion of the binary components
- The velocity V_e of the escaper with respect to the binary

The parameters Z, Ψ, V_b, V_e have been calculated at the time of the minimum moment of inertia. Here r_1 and r_2 represent the distances from the escaper to the binary components. In addition to the parameters describing the last triple approach, the final elements of the binary have also been determined, i.e. semi-major axis A and eccentricity E. We have adopted scaled units where masses are either equal to 1 or selected from the Pythagorean case (3,4,5). It is also useful to introduce the crossing time which is defined by $\tau = G(M_t)^{1/2}(M_1M_2 + M_1M_3 + M_2M_3)/(-2E)^{3/2}$ (Anosova, 1986), where M_t is the total mass. The velocity unit is then d/τ.

3. Statistics of Triple Approaches

Averages and r.m.s. deviations of the triple approach parameters are given separately in Table 1 for the regions D and R. For each region, we have studied 2500 different initial conditions. The corresponding distributions of parameters are given in Table 2, where the first and second line for each parameter refers to the regions D and R, respectively. Here Column 2 contains the range of the relevant parameters of Column 1. Then follow histograms for each region (truncated, in percent), with

TABLE I
Statistics of triple approach parameters.

Parameters	Region D	Circle R
T	72.6 ±2.6	79.5 ±1.4
P	0.655 ±0.012	0.746 ±0.010
I	0.469 ±0.046	0.271 ±0.006
Y	0.058 ±0.003	0.050 ±0.001
Z	0.332 ±0.010	0.330 ±0.007
Ψ	50.4° ±0.9°	51.9° ±0.7°
V_b	1.2 ±0.2	1.2 ±0.3
V_e	5.8 ±0.3	6.1 ±0.4
A	0.280 ±0.003	0.277 ±0.003
E	0.720 ±0.009	0.715 ±0.007

TABLE II
Distributions of triple approach parameters.

Param.	Range	Percentage											Median
T	0 – 100	23	12	10	6	6	4	4	3	2	2	21	40
		17	10	9	8	6	5	4	4	3	3	28	50
P	0 – 1.5	4	12	18	17	15	12	8	5	1	1	1	0.6
		3	8	11	15	17	14	10	8	4	3	4	0.7
I	0 – 1.0	27	20	15	11	7	3	1	2	1	2	4	0.22
		27	23	16	10	7	3	3	2	2	1	2	0.20
Y	0 – 0.1	23	12	10	8	6	6	4	3	4	2	17	0.04
		22	12	11	8	7	7	4	3	3	3	15	0.04
Z	0 – 1.0	24	19	13	9	7	6	6	4	4	4	0	0.25
		22	19	13	9	7	6	4	4	3	5	0	0.27
Ψ	0 – 90°	5	6	7	9	11	11	12	11	11	12	0	54°
		5	6	7	8	9	11	12	12	12	13	0	56°
V_b	−10 – 10	2	0	2	7	15	35	23	6	1	1	2	1.4
		1	0	2	8	14	35	25	6	2	0	2	1.4
V_e	0 – 10	0	1	15	29	19	10	6	3	2	2	9	4.2
		0	0	14	29	20	12	6	3	1	1	8	4.3
A	0 – 0.5	1	3	5	8	11	16	26	25	0	0	0	0.3
		0	3	6	8	13	16	23	25	0	0	0	0.3
E	0 – 1.0	1	3	4	5	5	7	9	10	13	37	0	0.8
		1	4	4	6	6	8	8	10	13	36	0	0.8
Theory		1	2	3	4	5	7	9	11	16	44	0	0.87

bin sizes equal to one-tenth of the range. The last entry of the histograms gives the proportion of triple systems having values exceeding the upper bound. Finally, the last column shows the median values of the distributions in appropriate units; i.e. T is in units of the crossing time, etc.

The statistical results of the tables are essentially similar for the regions D and R, confirming earlier work (Anosova *et al.*, 1993). Some specific features can be summarized as follows:

- The distribution of the time T has a peak within the interval (0, 10).
- The distribution of the perimeter P has a flat maximum within (0.3, 0.75) for the region D and (0.45, 0.9) for the circle R, whereas the medians are similar and $\simeq 2d/3$. We note that the medians of A are a factor of 2 smaller, while the average distance between the bodies at the triple approach is about $d/4$.
- The moment of inertia of the triple systems has a distribution with a peak in the interval (0, 0.1) and a median $\simeq 0.2$ (in units of mass and d^2 such that $I = 1$ in virial equilibrium). Thus the distributions of P and I confirm the earlier results (see, e.g., Anosova and Zavalov, 1981; Anosova, 1991) that the triple approach leading to escape is usually a close one.
- The distribution of the impact parameter has a peak near zero: about a quarter of all values of Y are in the interval (0, 0.01). Hence in these triple approaches, the passages of the escaper are preferentially near the centre of mass. Before the last triple approach, the escaper usually experiences a long ejection, and therefore gravitational focusing will tend to direct its orbit towards the centre of mass of the binary.
- Cases close to the isosceles triangle with $Z \in (0, 0.3)$ predominate; hence the configuration tends to be symmetrical at the time of the closest approach. A separate study of such deviations based on a total of 2500 random samples inside a circle show that this distribution is somewhat flatter (with corresponding bins 17, 16, 14, 14, 12, 10, 8, 5, 4, 1) when determining Z, where the maximum distance represents the binary components as in the dynamical case.
- The angle Ψ is more often close to 90° than to 0, such that about one-third of the cases have $\Psi < 45°$ and two-thirds have $\Psi > 45°$. The distribution of Ψ within (45, 90°) is approximately flat. Thus the trend is for the direction of the motion of the escaper to be nearly orthogonal to the line connecting the two other components.
- The binary components are more often ($\simeq 3/4$) receding at the closest approach. In the opposite case, an exchange triple approach of type (II) may take place.
- The velocity of the escaper at the close triple approach is rather large, typically four times the virial velocity and in almost 10 % of all cases, ten times this value. Note the absence of triple approaches with small escaper velocity.

TABLE III
Comparison of statistics for different problems

Parameters	Regions $D + R$	Problem $\mathcal{P}$
T	76.0 ±1.7	52.4 ±0.8
P	0.700 ±0.008	0.805 ±0.018
Y	0.054 ±0.001	0.051 ±0.007
Z	0.331 ±0.006	0.213 ±0.028
Ψ	51.2° ±0.6°	60.4° ±2.8°
V_b	1.2 ±0.2	2.0 ±0.4
V_e	6.0 ±0.2	4.9 ±0.5
A	0.278 ±0.002	0.272 ±0.008
E	0.718 ±0.006	0.858 ±0.018

- The final binaries have a distribution with a sharp cutoff at $A \simeq 0.4$. This is connected with an assumed value of $10\,d$ for the conditional escape distance. Thus if the binary is wider, the ejected body will return from a distance $< 10\,d$ for further interactions.
- The eccentricity distributions $f(E)$ have a peak in the interval (0.9, 1) and resemble the theoretical one, $f_t(E) = E/(1 - E^2)^{1/2}$, derived by Monaghan (1976) from statistical considerations of the escape process in planar triple systems (see last line of Table 2).

The above features are intrinsic to triple systems with equal-mass components. In order to verify the generality of these features, we have considered triple approaches resulting in escape for unequal masses based on the Pythagorean problem.

A comparison of the statistics of triple approaches for the regions D and R (5000 cases) versus the Pythagorean problem $\mathcal{P}$ is given in Table 3. The symbols here are the same as in Table 1. For the problem $\mathcal{P}$ only 831 triple systems with predictable orbits have been included (see Aarseth *et al.*, 1994). Here the parameters have been re-scaled to the same units as used for equal masses. It can be seen that most results are in agreement; the longer life-times for equal masses are partly due to mass effects and the fact that non-predictable cases are included. The general features of triple approaches for unequal masses can be summarized as follows:

- The average close approach of the escaper to the centre of mass is $<Y> \simeq 0.05$
- The average closeness to the isosceles triangle is $<Z> \simeq 0.2$, similar to results obtained from random configurations which are mainly non-hierarchical
- The radial velocity of the binary components at the close triple approach is positive in $\simeq 90\%$ of the cases
- The velocity of the escaper is large: $<V_e> \simeq 5$
- The velocity vector of the escaper tends to be nearly orthogonal to the line connecting the binary components: $<\Psi> \simeq 60°$

It should be noted that in systems with zero angular momentum, the escaping body can only pass near the centre of mass if the binary eccentricity is very large since the angular momentum of the binary itself must be counterbalanced by that of the escaper. Table 2 shows that more than one-third of all cases have final eccentricities exceeding 0.9, whereas the average eccentricity for the Pythagorean problem is also large ($\simeq$ 0.86). As far as the general classification scheme is concerned, the states 4 and 5 are not realized in the examples studied here. However, states of zero total angular momentum can give rise to stable revolution if the outer body has a retrograde orbit with sufficiently large pericentre.

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References

Aarseth, S.J., Anosova, J.P., Orlov, V.V. and Szebehely, V.G.: 1994, 'Global chaoticity in the Pythagorian three-body problem', *Celest. Mech.*, **58**, 1–16.

Agekian, T.A. and Anosova, J.P.: 1967, *Sov. Astron. Zh.*, **44**, 1261.

Agekian, T.A. and Anosova, J.P.: 199,. *Sov. Astron. Zh.*, **68**, 1099.

Agekian, T.A. and Martynova, A.I.: 1973, *Messenger Leningrad Univ.*, **1**, 122.

Anosova, J.P.: 1986, *Astrophys. Space Sci.*, **124**, 217.

Anosova, J.P.: 1988, In *The Few Body Problem*, p. 27, ed. M.J. Valtonen, Kluwer Academic Publishers, Dordrecht.

Anosova, J.P.: 1991, 'Strong Triple Interactions in the General Three-Body Problem', *Celest. Mech.*, **51**, 1–15.

Anosova, J.P. and Orlov, V.V.: 1992, *Astron. Astrophys.*, **260**, 473.

Anosova, J.P., Orlov, V.V. and Aarseth, S.J.: 1993, (submitted, *Celest. Mech.*).

Anosova, J.P. and Zavalov, N.N.: 1981, *Publ. Astron. Obs. Leningrad Univ.*, **36**, 109.

Anosova, J.P. and Zavalov, N.N.: 1989, *Sov. Astron. Zh.*, **66**, 152.

Golubev, V.G. and Grebenikov, E.A.: 1985, *The Three-Body Problem in Celestial Mechanics*, Moscow State University.

Marchal, C.: 1990, *The Three-Body Problem*, Elsevier Publishers, Amsterdam.

Monaghan, J.J.: 1976, *Mon. Not. R. astr. Soc.*, **176**, 63.

Orlov, V.V.: 1986, *Messenger Leningrad Univ.*, **1**, 82.

Standish, E.M.: 1971, 'Sufficient Conditions for Escape in the Two-Body Problem', *Celest. Mech.*, **4**, 44–48.

Szebehely, V.: 1971, 'Classification of the Motions of Three Bodies in a Planet', *Celest. Mech.*, **4**, 116–118.

Szebehely, V., 1979. *Astron. Astrophys.*, **78**, 349.